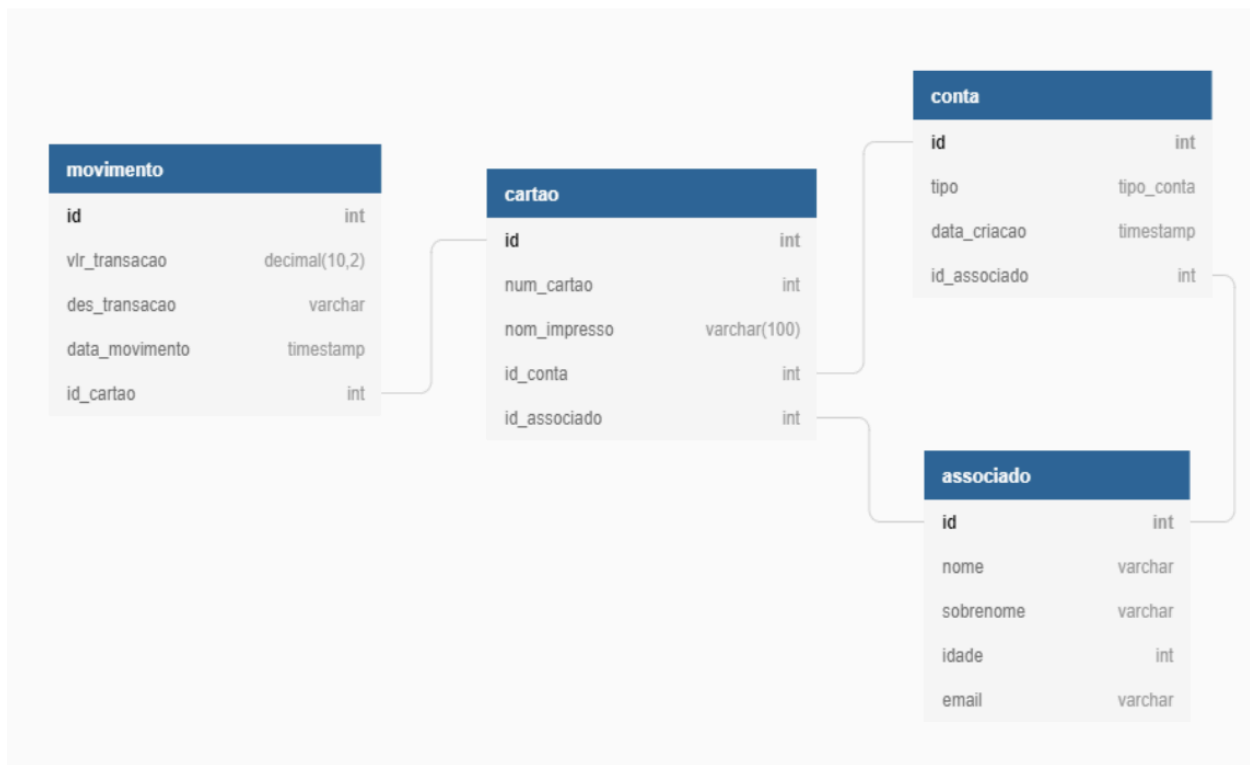


Technical challenge data engineering

Problem:

Let's say you're part of the data engineering team at a company called SiCooperative LTDA, whose purpose is to offer financial solutions that are fair to people. However, a major problem was impacting the speed and assertiveness of decision-making by managers and directors. This problem can be explained mainly by the difficulty in aggregating different pieces of information into a single point; a lot of time was being wasted on creating individual reports and then trying to correlate them manually. In addition, a new Data Science team was being set up with the aim of creating predictive models that would have the ability to analyze the current moment of each associate (client), and thus offer solutions that are more coherent with each member's reality. After a strategy meeting, it was decided to start a POC to create a Data Lake, with the aim of solving the aforementioned problems. After this, an initial mapping exercise was carried out to define the ingestion priorities, i.e, what would be the first information to be brought into the new structure, and as a result the card movement data was chosen. It contains information on every transaction made by members' cards, with information on the value and date of the transaction. In addition, the tables to which it relates must also be loaded: the card table, which contains additional card information, the account to which that card and member are linked, and member information. A member can have several cards and several accounts.

The diagram of this structure is organized as follows:



Challenge:

You must first model the structure described above in a database of your choice, then create a component capable of reading the data from the tables created and writing it to a single flat file in a local directory. a single apartment file, in a local directory, i.e. all the tables must be aggregated to form a single view, where the structure of the file is represented as follows:

movimento_flat	
nome_associado	string
sobrenome_associado	string
idade_associado	string
vlr_transacao_movimento	string
des_transacao_movimento	string
data_movimento	string
numero_cartao	string
nome_impresso_cartao	string
data_criacao_cartao	string
tipo_conta	string
data_criacao_conta	string

Requirements:

- Create the structure of the database to be ingested, using the technology of your choice (MySQL, PostgreSQL).
- Insert a mass of fictitious data into the tables, it doesn't have to be such a large volume.
- Use the programming language of your choice.
- In order to ETL the data, a distributed processing framework for Big Data must be used, e.g. Hadoop, Spark, Flink and Storm.
- Write a CSV file, in a directory parameterized by the user.
- Make this project available in a private repository on your GitHub.

Bonus:

- BonusProvision the environment and carry out an initial execution of the process in an automated way using Docker.
- Create unit tests to cover the main parts of the flow.

Observations:

When you create your project, you can document everything in the README: why you chose a particular design, what you would do if you had more time to complete the challenge, any difficulties you encountered during development. It is important to note that all the scripts used to create the structures must be saved within your project.